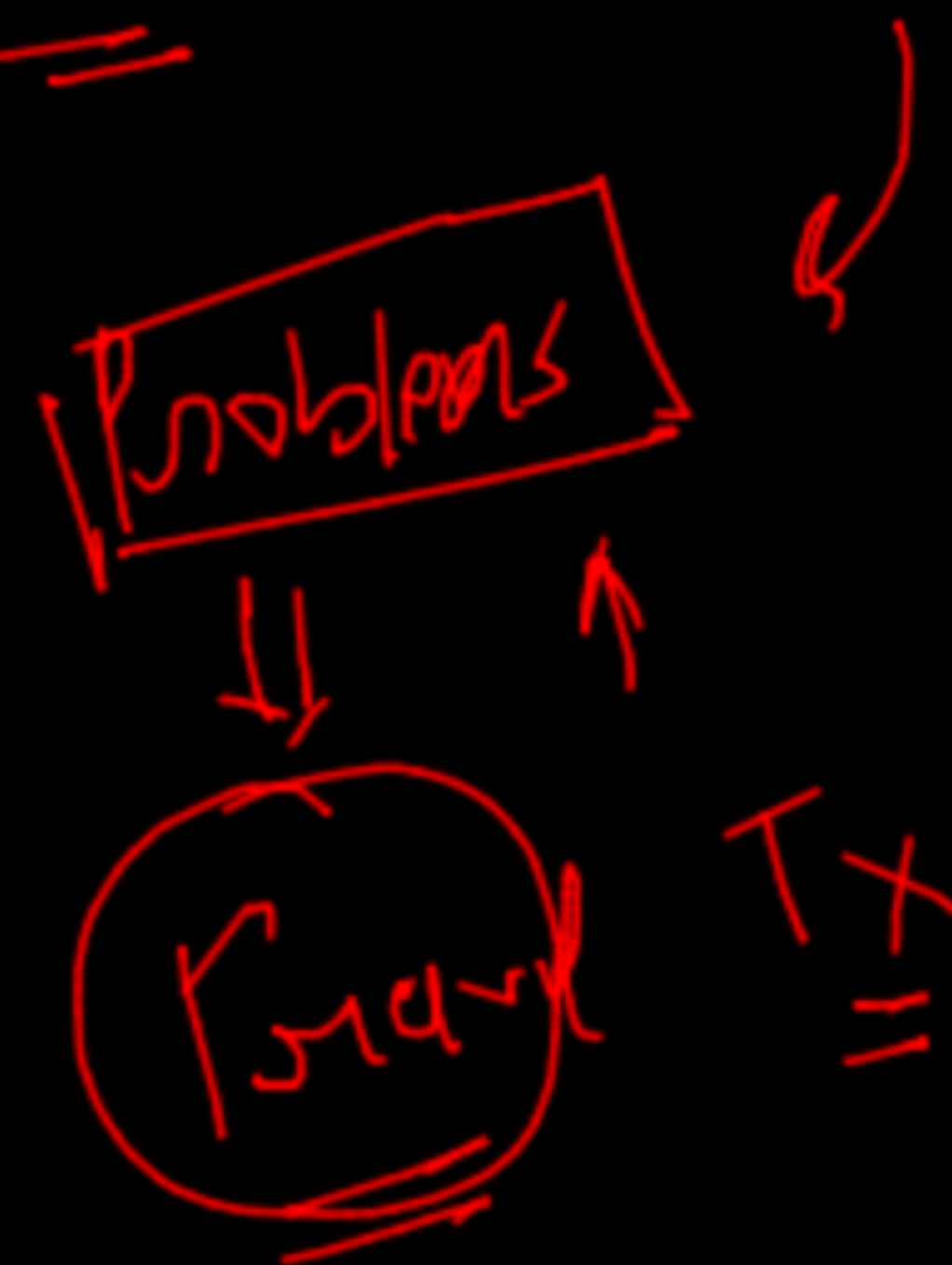


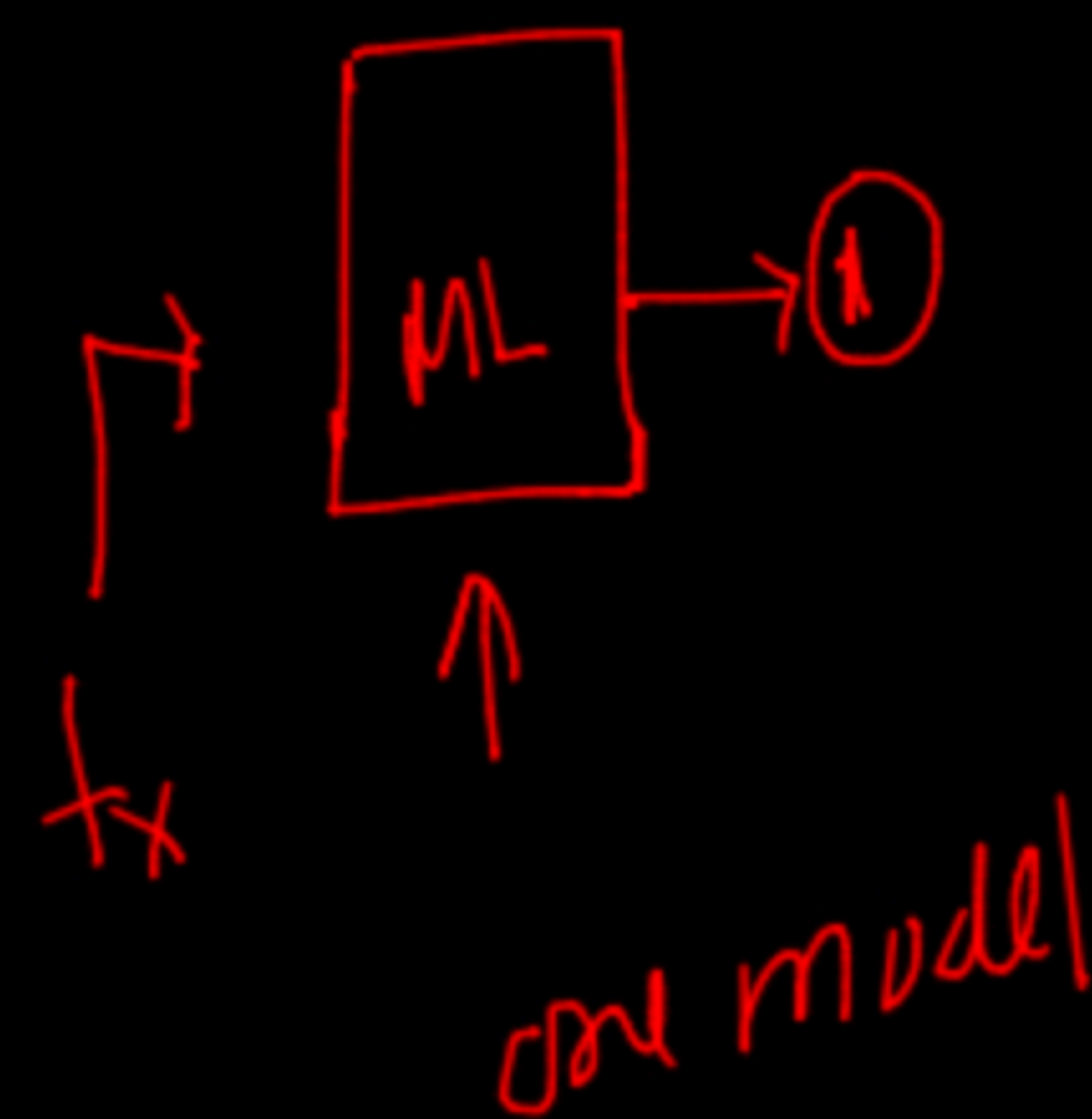
Ensemble

→ 911

Group of things



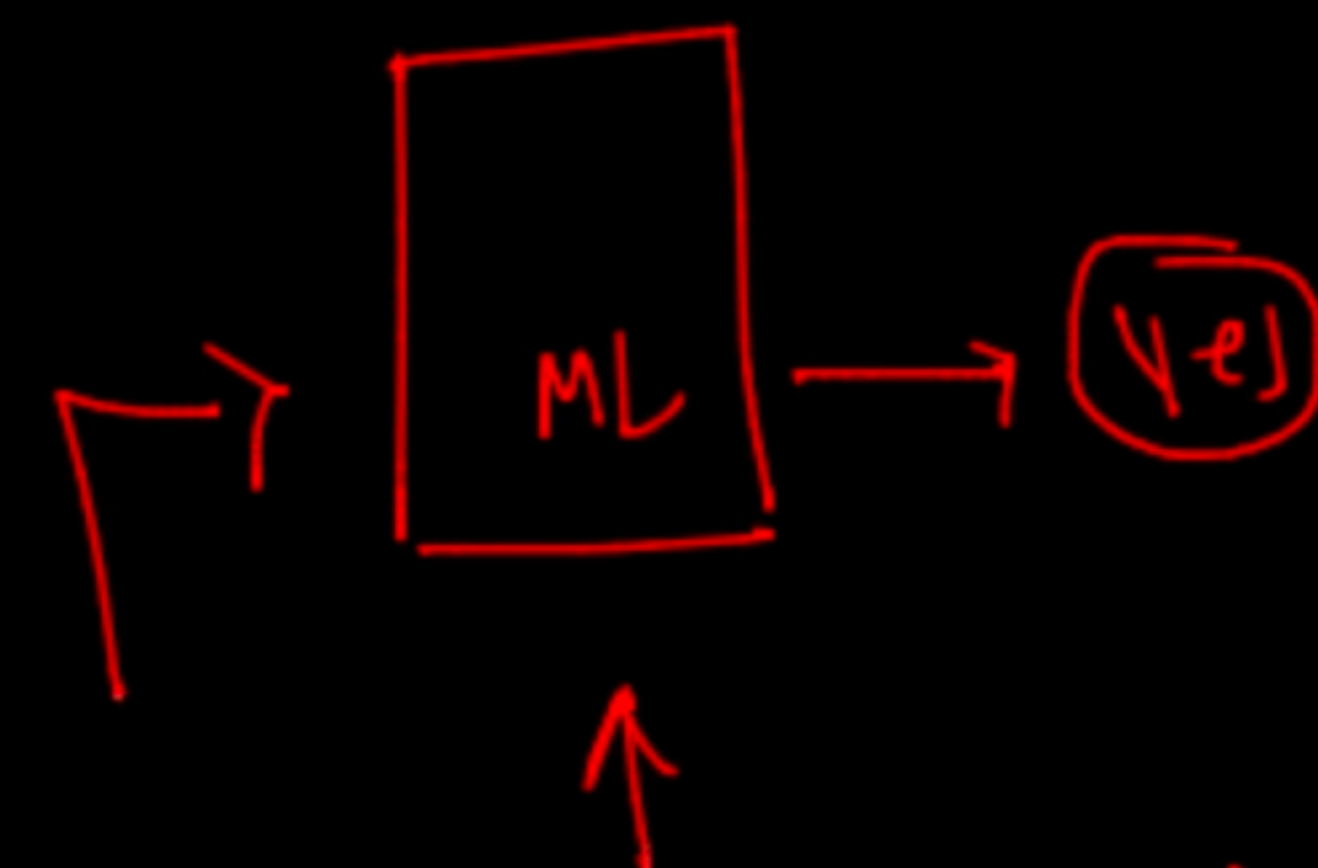
$T_x \rightarrow ?$



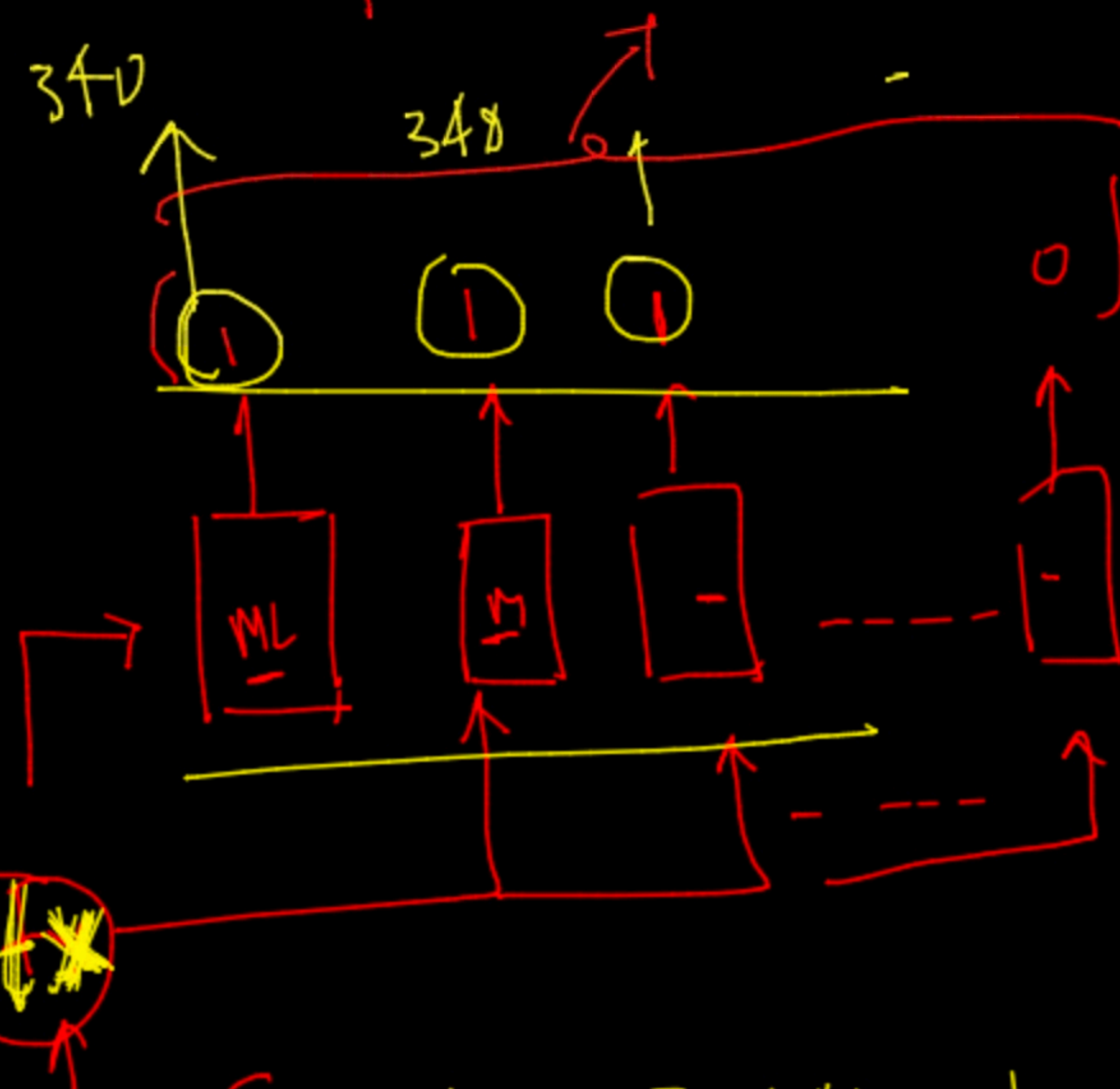
941

94, 91

901
10



10 → 6 ✓
4 ✓
11 ✓



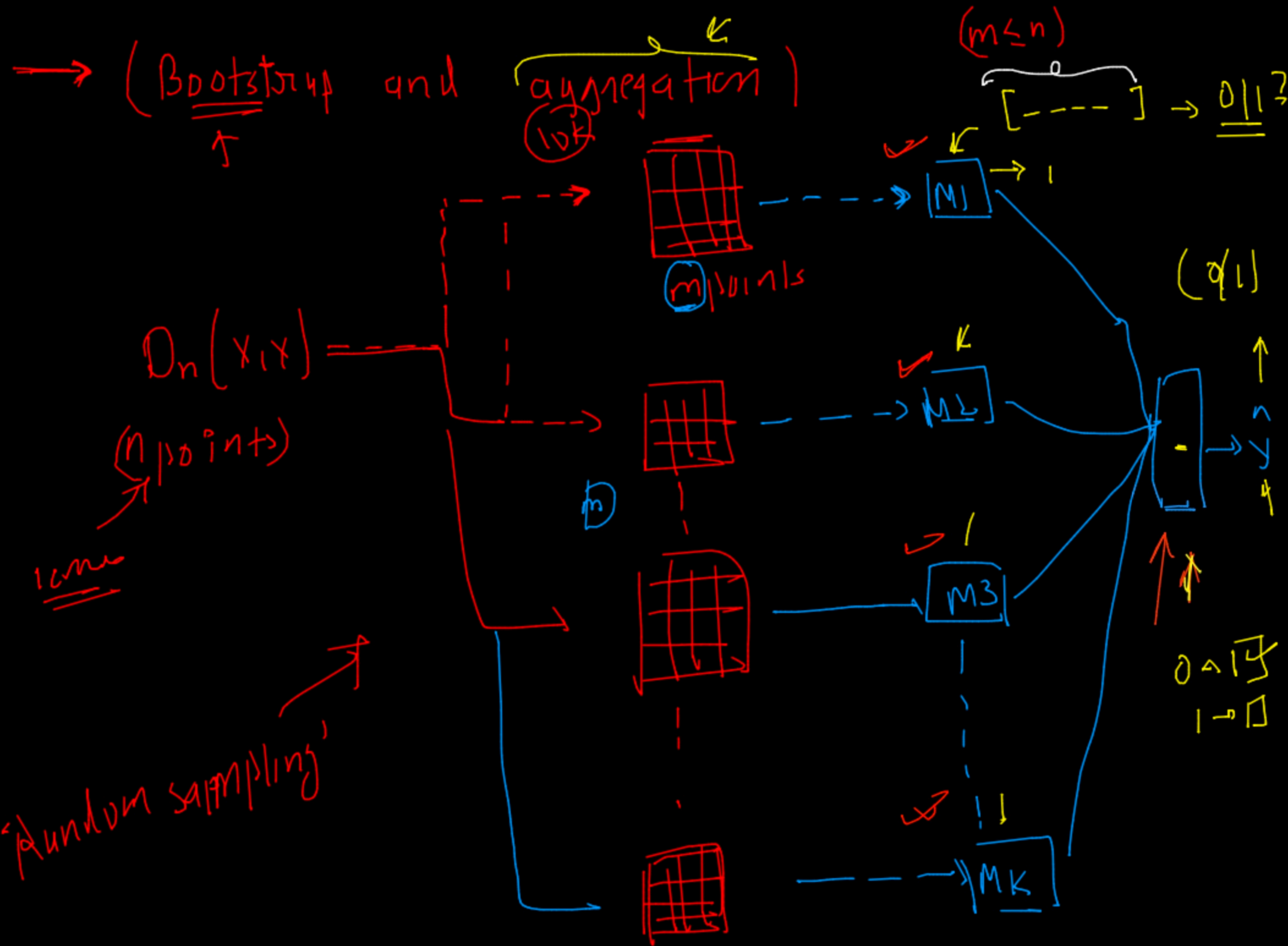
'depression'

$$F = \frac{348 + 341 + \dots}{5}$$

- ① Bagging imp
- ② boosting imp

③ stacking

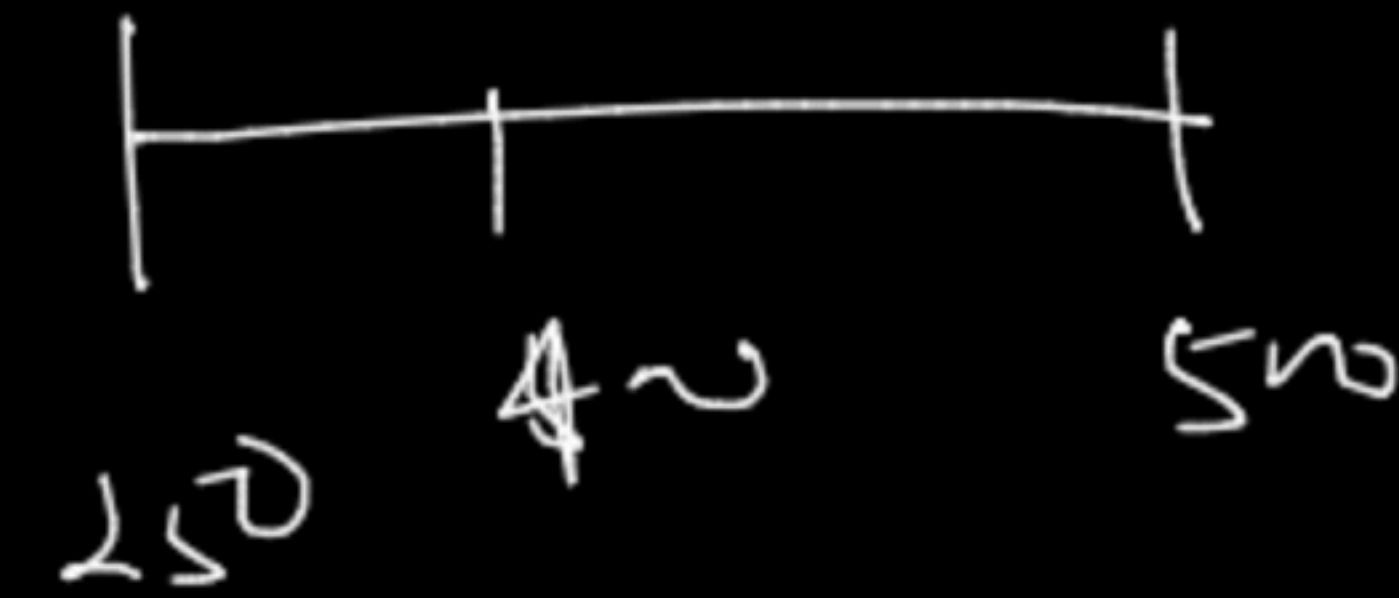
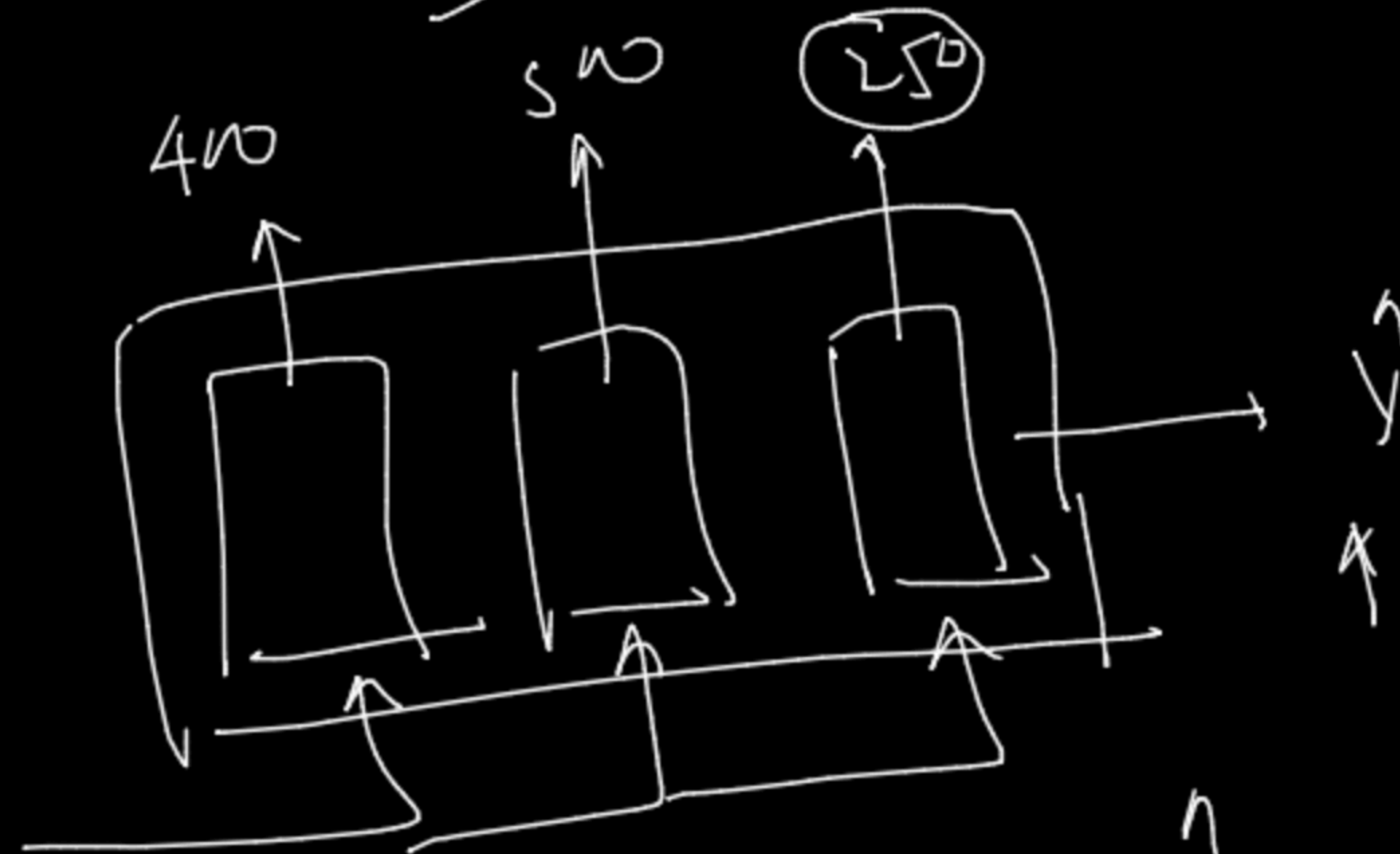
④ cascading



Aggregation

① In classification $\rightarrow$ majority voting

② In Regression $\rightarrow$ mean | median



$$\frac{400 + 500 + 250}{3}$$

$$\hat{y} = \frac{400 + 500 + 250}{3}$$

$$\Rightarrow \frac{380}{3}$$

(--)

Bagging

low bias
+
high variance

Take bunch of low-bias + high variance model (DT)

Combine using Bagging

Follows

train
test
781.7
181
20.1

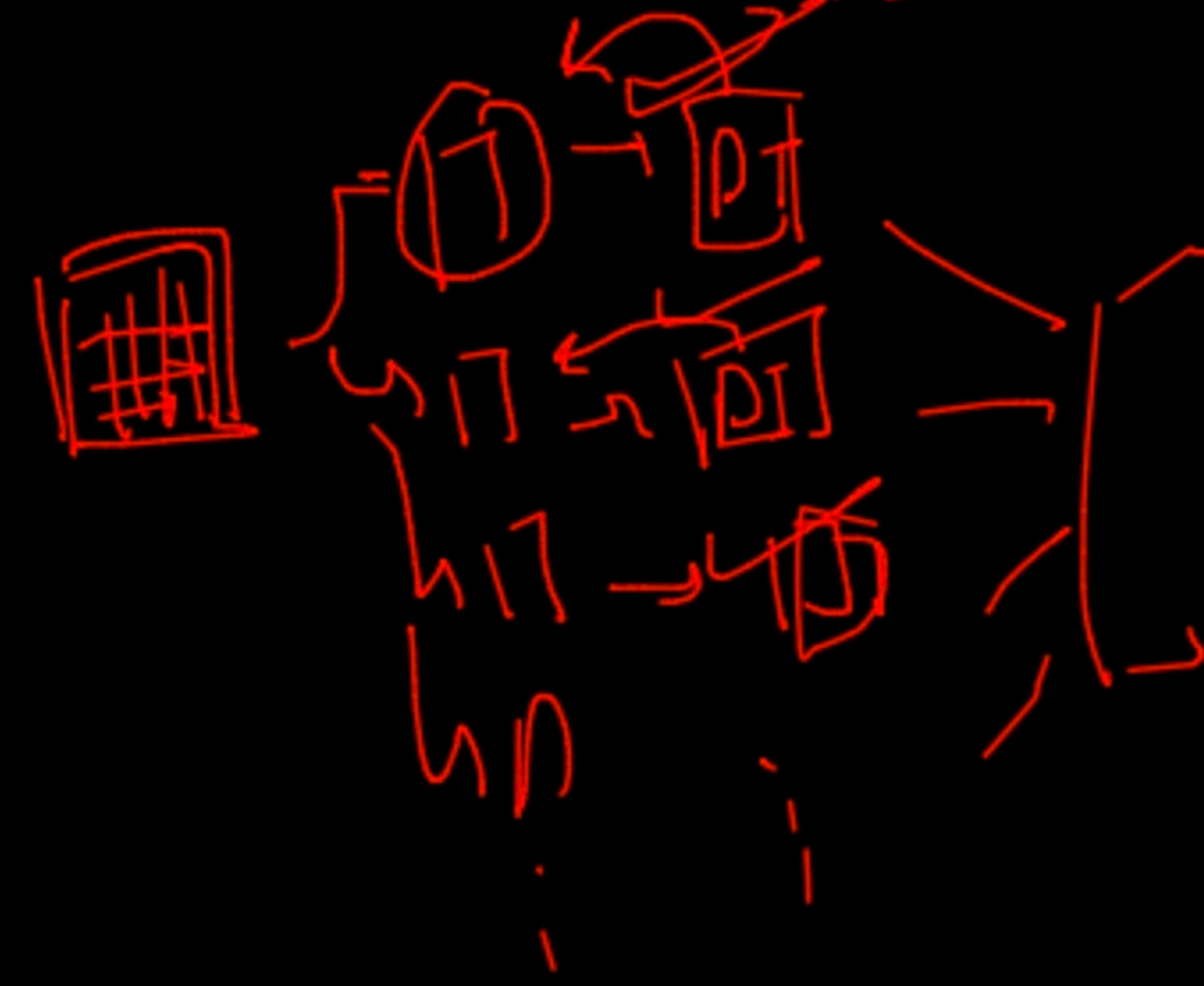
Algorithm

Random

Forest (100)

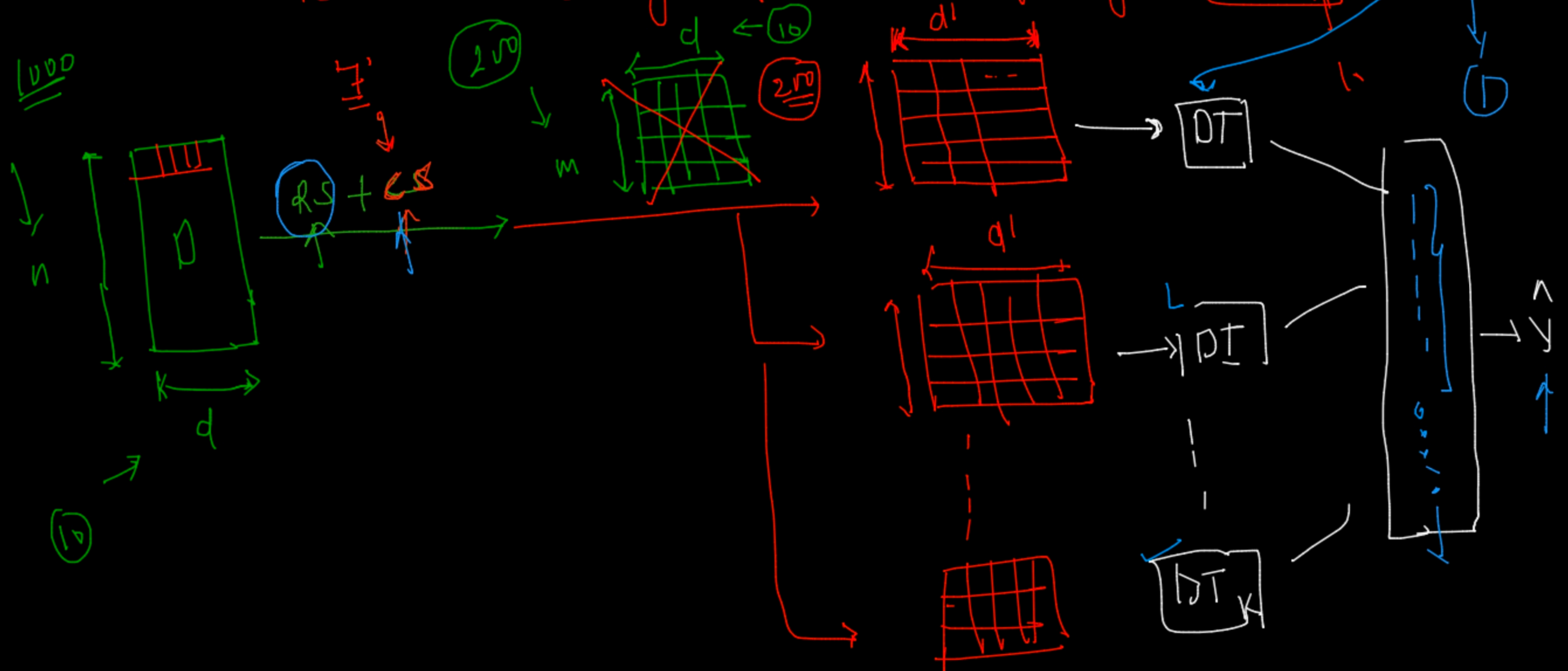
G.S. 12
R.S. 2

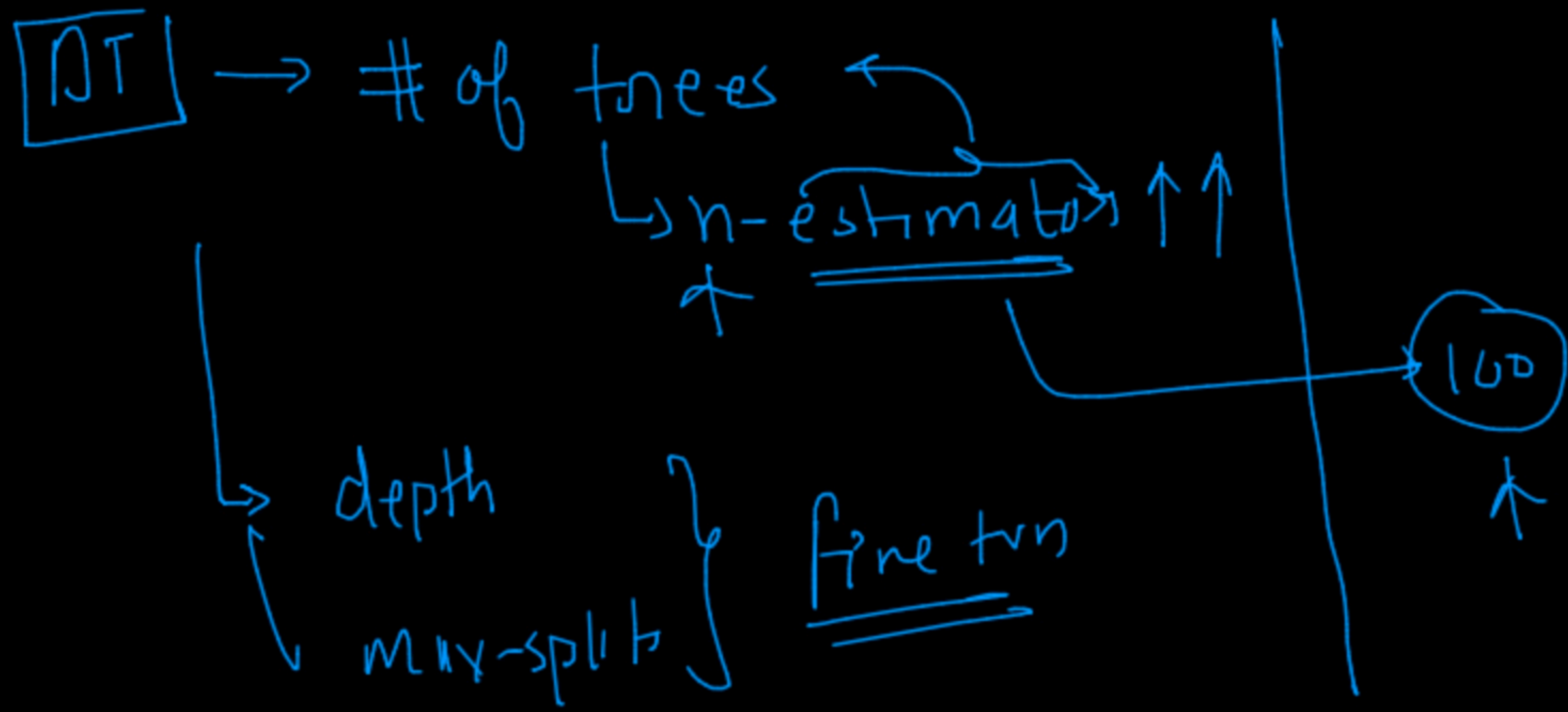
(Low Bias + Low Variance)



RF $\Rightarrow$ will take DT as base learner and will apply Bagging and col sampling

RF $\Rightarrow$ DT + Bagging + col. sampling





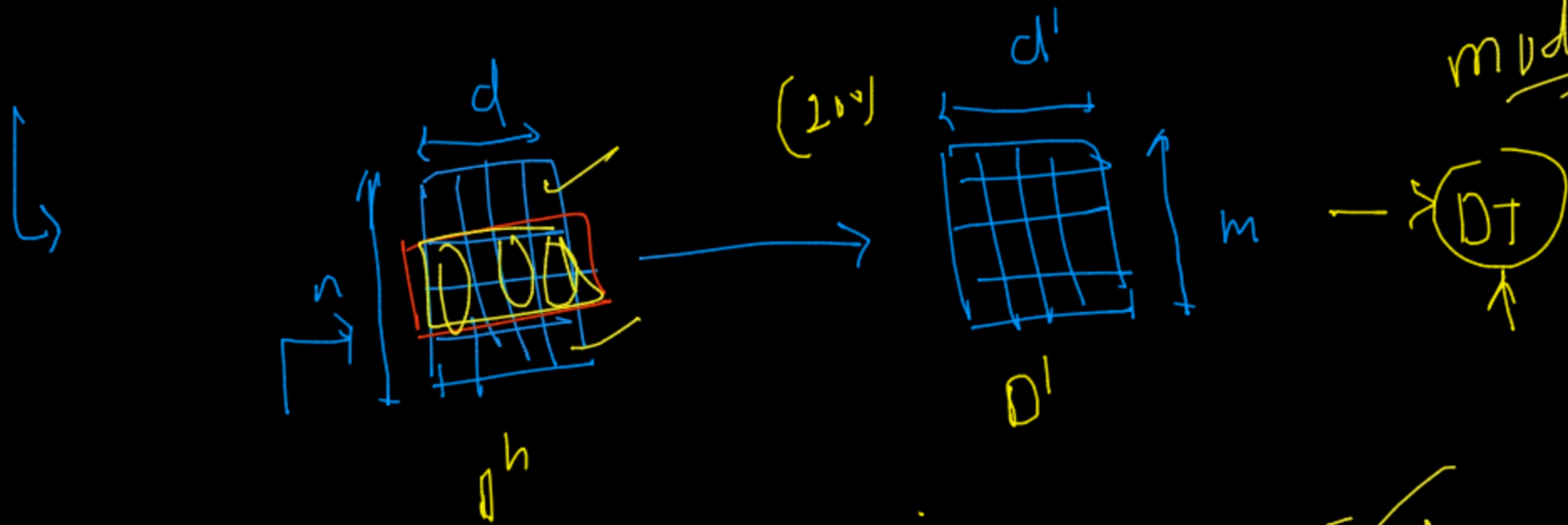
Split

$$\text{model} = \text{DT}(X_{\text{tr}})$$

$$\text{model.predict}(X_{\text{test}})$$

18%

(2%)



$(D^h - D')$

└─ (testing)

→ oob = True

(bag of points)

Bias-variance tradeoff

Random forest → reduce variance

→ $M = \text{agg}(M_1, M_2, M_3, \dots, M_K)$

$K \uparrow$ variance $\downarrow$

$K \downarrow$ variance $\uparrow$

case $K=1 \rightarrow DT$ — High var

